

13.2 Compound Interest

Lesson Introduction

 Benchmarks	MA.912.FL.3.2 Solve real-world problems involving simple, compound, and continuously compounded interest.		 Learning Targets	- I can explain the relationship between compound interest and exponential growth. - I can solve real-world problems involving compound interest. - I can compare simple interest growth and compound interest growth.
MA.912.FL.3.4 Explain the relationship between simple interest and linear growth. Explain the relationship between compound interest and exponential growth and the relationship between continuously compounded interest and exponential growth.				
MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.				
 Benchmark Coherence	Building On	Working Toward		
	N/A	MA.912.FL.3.1 Compare simple, compound, and continuously compounded interest over time. MA.912.FL.3.4 Goes beyond clarification where exponential growth is limited to compound interest.		
 Essential Questions	- What is the relationship between compound interest and exponential growth? - How are real-world compound interest problems solved?			
 Lesson Narrative	In this lesson, students are introduced to the concept of compound interest. They begin by learning about what compound interest is and how it is calculated. Students will connect this concept to that of exponential growth that they have learned in prior lessons. They will transition into a comparison between financial situations involving simple vs compound interest.			
 Component Summary	13.2.1 Warm-Up (~5 min.)	Growth mindset reflection through an interview with a paramedic (<i>SE p. 241</i>)	13.2.3 Guided Instruction (15-20 min.)	Introduction to calculating compound interest (<i>SE p. 243</i>)
	13.2.2 Collaboration (10-15 min.)	Understanding compound interest (<i>SE p. 242</i>)	13.2.4 Collaboration (10-15 min.)	Comparing interest methods (<i>SE p. 245</i>)
	13.2.5 Wrap-Up (~5 min.)	Ticket Out the Door on comparing interest methods (<i>SE p. 246</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Compound interest <u>Additional Terms:</u> N/A		(Optional) Warm-Up video	N/A

 Teacher Tips	Common Misconceptions - Students may provide incorrect values for compound interest. For example, students may read “quarterly” and write 25, or students may read bi-monthly and write 2 (“bi” for 2). - Students may distribute the coefficient P value (Principal) before raising the quantity to the power nt.	Things to Consider Activate prior knowledge in 13.2.3 Guided Instruction, making connections to content from Unit 11 with exponential functions, their graphs, and key features. This also means that Math Nation Unit 11 resources can be utilized to provide additional scaffolding and remediation during this lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 13.2.3 Guided Instruction introduces students to the concept of calculating compound interest. While this concept is new, it is designed to reinforce skills learned in Unit 11 with exponential functions and graphs. During this activity, allow students to hear your thinking out loud as you connect the prior learning to the current learning.	MA.K12.MTR.5.1: Patterns and Structure One of the purposes of this lesson is to connect a new concept to what students have learned in prior units. What students are asked to do in the context of compound interest connects to their learning with exponential functions. For this lesson, students will be relating previously learned concepts to new concepts and looking for similarities among problems.
 Differentiate Instruction	When dealing with financial situations, the level of student interest and engagement tends to be higher because of their ability to connect with the content. Consider leveraging this to help students understand the material. For example, consider differentiating 13.2.4 Collaboration to allow student choice. Have students find investment opportunities as two different locations that meet the same conditions as the table and have them complete it on their findings. Another alternative is to have students verify their answers using free online investment calculators to explore the investment over their chosen intervals of time.	
Lesson Notes:		

13.2.1 Warm-Up: Paramedic

Component Overview: Students will explore the career of a paramedic and answer two questions related to staying focused and learning difficult concepts in mathematics.



13.2.1 Warm-Up: Paramedic

1. Kelly Grayson is a paramedic, or emergency medical technician (EMT). In an interview, she was asked, "How do you stay focused and deal with the crisis situation to solve the problem at hand when responding to an emergency call?" Kelly Grayson replied, "You focus on process and fall back on your training."

- a. How does staying focused on the process and falling back on your prior training help you in difficult situations?

Answers will vary. Take what you learn from previous concepts in math or lessons and apply it to the problem at hand that you are working on.

- b. What is one thing you have done to help you when learning difficult concepts in mathematics?

Answers will vary. Take parts of the concept and break it apart to relate it to concepts that I have previously learned. Find key parts of the problem.



The video for this Warm-Up is optional. If playing the video, consider having the video loaded and prepared prior to students entering the class to avoid any potential technical issues.

13.2.2 Collaboration: Understanding Compound Interest

Component Overview: Students will work with a partner or small group to complete three questions in the process of exploring compound interest.



13.2.2 Collaboration: Understanding Compound Interest

Compound interest is a method of computing interest. Interest is computed from the (original) principal and the interest earned during a certain time period.

The formula $A = P \left(1 + \frac{r}{n}\right)^{nt}$ is used to calculate the final amount owed or saved, where A = final amount, P = principal or original amount, t = number of years, n = number times per year the interest is being compounded, and r = rate of interest per year.

When money is invested, compound interest means that interest is being accrued based on both the principal and previous interest earned.

When money is owed, such as a credit card bill, compound interest means that interest is owed based on both the principal and previous interest accrued.

1. Jasmyne invests \$10,000 into a savings account with an interest rate of 6% annual percentage rate (APR) that is compounded quarterly. The table shows the amount of Jasmyne's savings for different compounding periods after 1 and 10 years.

Compounding Period	Amount After 1 Year	Amount After 10 Years
quarterly	\$10,613.64	\$18,140.18
monthly	\$10,616.78	\$18,193.97
daily	\$10,618.31	\$18,220.29



Students are likely to not be familiar with the term “compound interest.” Leverage the narrative language underneath the definition to help provide additional support for understanding this concept. This narrative language provides an entry point for a discussion about how it can be good and bad in general terms.

13.2.2 Collaboration: Understanding Compound Interest (continued)

- a. What do you notice about the table?

Answers will vary. I notice that when it is compounding daily, the amount is higher. I notice that she makes more interest when it is compounded more frequently.

- b. Discuss with your partner why the amount after 1 year is different for each compounding period. Summarize your discussion.

Answers will vary. They are all compounding at different periods of time.

2. Work with your partner to explain why compound interest for investments, such as savings accounts or certificates of deposits (CD), are better than simple interest.

There is more interest earned, which can result in a higher amount of money in the accounts in the future that someone can have.

3. Work with your partner to explain why compound interest for loans, such as a car loan, is not better than simple interest.

With simple interest, the interest occurs on the overall amount. With compound interest, there is interest being added at each time period. With that interest being added into the amount, it makes the interest in the next time period based on the original plus the added interest from the previous time period. This will result in a much larger amount to owe.



The conversation around the comparison between this concept and simple interest is very important. Once the time for this activity is over, spark a conversation with the students about the things they discovered during this activity and relate it to their prior learning with simple interest.



Consider providing student choice for engaging in question 3, giving students the opportunity to make sense of compound interest through their lived worlds.

Prompt students to give an example using mathematics to prove their reasoning.

- Car Loan
- Student Loan
- Mortgage (Home Loan)
- Student Debt
- Credit Card Debt
- Create Your Own

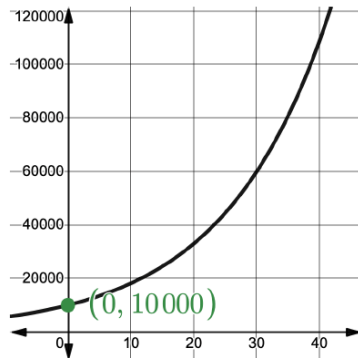
13.2.3 Guided Instruction: Compound Interest

Component Overview: Students will be introduced to calculating compound interest and the relationship between compound interest and exponential functions, their graphs, and key features.



13.2.3 Guided Instruction: Compound Interest

1. The graph shows an initial investment of \$10,000 in a savings account. The interest rate is 6% compounded quarterly.



- a. What do you notice about the graph?
Answers will vary. I notice that it is a curved graph that is increasing. I notice that the graph appears to be exponential.
- b. What type of function best represents the graph?
☐ linear
☐ quadratic
☒ exponential
- c. Discuss with your partner how the formula for compound interest, $A = P \left(1 + \frac{r}{n}\right)^{nt}$, supports your function choice. Summarize your discussion.
There is an exponent in this formula, resulting in an exponential function.
- d. Complete the table.

Key Feature	Description
y-intercept	\$10,000 or (0,10000)
End Behavior	As $x \rightarrow \infty$, $y \rightarrow \infty$.
Increasing/Decreasing	As x increases, $y \rightarrow$ <i>increases</i> .



During this activity, discuss with students their answers to each of these questions.

- Discuss with your partner why the amount after one year is different for each compounding period. Summarize your discussion.
- Work with your partner to explain why compound interest for investments such as savings accounts or certificates of deposits (CDs) are better than simple interest.
- Work with your partner to explain why compound interest for loans is not better than simple interest.



For part d, students can give the ordered pair or the value itself. If specific formats are required, that information should be provided verbally. Mathematically, either are appropriate.

13.2.3 Guided Instruction: Compound Interest (continued)

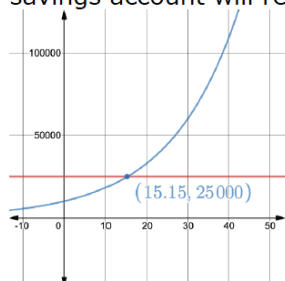
- e. Use the information about the graph to write an equation that models the relationship.

$$A = 10,000 \left(1 + \frac{0.06}{4} \right)^{4t}$$

- f. What is the balance of the savings account after 7 years?

\$15,172.22

- g. Use the graph shown to determine when the savings account will reach \$25,000.



15 years, and 54 days

2. Amir deposited \$32,500 in a savings account with a 4.25% APR compounded monthly.

- a. Write the equation that models the amount of money Amir has in the account.

$$A = 32500 \left(1 + \frac{0.0425}{12} \right)^{12t}$$



Utilize inquiry to promote meaningful discourse and promote MTR 4.

- What are some things that you notice about the graph in question 1g?
- How does the red line help you determine when the savings account will reach \$25,000?



Consider guiding students through part a of question 2 and then release students to work on part b on their own and work with their partner for question 2c.

13.2.3 Guided Instruction: Compound Interest (continued)

- b. Use the equation to complete the table. Show your work.

Number of Years	Total Amount Saved
5	$A = 32500 \left(1 + \frac{0.0425}{12} \right)^{12(5)}$ $A = 32500 \left(1 + \frac{0.0425}{12} \right)^{60}$ $A = 32500 \left(\frac{4817}{4800} \right)^{60}$ $A = \$40,179.81$
35	$A = 32500 \left(1 + \frac{0.0425}{12} \right)^{12(35)}$ $A = 32500 \left(1 + \frac{0.0425}{12} \right)^{420}$ $A = 32500 \left(\frac{4817}{4800} \right)^{420}$ $A = \$143,468.02$



Students may make calculation errors during their computations. Consider allowing students the ability to use calculators as part of this process.

- c. Discuss with your partner the relationship between compound interest and exponential growth. Summarize your discussion.
- Answers will vary. Exponential growth occurs from a percentage or rate of increase. Compound interest occurs similarly due to interest being added on top of the previous principal and interest that has already occurred.*

13.2.4 Collaboration: Comparing Interest

Component Overview: Students will work with a partner to compare an investment that has been invested using simple and compound interest scenarios.



13.2.4 Collaboration: Comparing Interest

1. Work with your partner to complete the table to compare two investments.

	Simple Interest	Compound Interest
Investment amount and APR	\$50,000 is invested at an interest rate of 2.75% APR.	\$50,000 is invested at an interest rate of 2.75% APR compounded monthly.
Write the equation.	$A = P(1 + rt)$ $A = 50000(1 + 0.0275t)$	$A = P\left(1 + \frac{r}{n}\right)^{nt}$ $A = 50000\left(1 + \frac{0.0275}{12}\right)^{12t}$
Determine the amount after 10 years.	$A = 50000(1 + 0.0275(10))$ \$63,750	$A = 50000\left(1 + \frac{0.0275}{12}\right)^{12(10)}$ \$65,805.83
Determine the amount after 20 years.	$A = 50000(1 + 0.0275(20))$ \$77,500	$A = 50000\left(1 + \frac{0.0275}{12}\right)^{12(20)}$ \$86,608.14



When circulating during this activity, look for the values that students are using to evaluate the expression. If students are inputting the wrong values, even if they do the calculations correctly, they will end up with the wrong answer. Try to cut this off by viewing the input values students are using during this activity.



Why does the compound interest plan give more money after 20 years? Is there a time frame where the simple interest would provide the most money?

13.2.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door by comparing an investment made in an account using simple interest vs compound interest. Students should be allowed to access a calculator to complete the Wrap-Up.



13.2.5 Wrap-Up: Ticket Out the Door

1. Complete the table to compare two investments.

	Simple Interest	Compound Interest
Investment amount and APR	\$2,325 is invested at an interest rate of 4.5% APR.	\$2,325 is invested at an interest rate of 4.5% APR compounded monthly.
Write the equation.	$A = P(1 + rt)$ $A = 2325(1 + 0.045t)$	$A = P \left(1 + \frac{r}{n}\right)^{nt}$ $A = 2325 \left(1 + \frac{0.045}{12}\right)^{12t}$
Determine the amount after 8 years.	$A = 2325(1 + 0.045(8))$ $= 3162$	$A = 2325 \left(1 + \frac{0.045}{12}\right)^{12(8)}$ $= 3330.25$



Consider “bonus” or “challenge” questions for students to attempt if time permits.

Would the simple interest ever produce more money than compound interest? Why or why not? Provide an example or counterexample.